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Paper Code : ESCS201 Programming for Problem Solving

UPID : 002008

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) Point out the difference between strcmp() and strncmp()
- (ii) What do you mean by pointer arithmetic in C?
- (iii) Write the main of the differences between Structure and Union?
- (iv) Find the correct output:

```
#include<stdio.h>
main()
{
for(int i=0; i<=5;i++);
{
printf("%d",i);
}
}
```
- (v) What is the main difference between call by value and call by address in C?
- (vi) Why is scope of variable necessary in function?
- (vii) #include

```
main()
{
int i=3;
i=i++;
printf("%d",i);
}
```

What will be the output of the above code?
- (viii) Why do we use header files?
- (ix) Write the output of the following Code:

```
#include
int main()
{
char x;
x = 'a';
printf("%d\n", x);
}
```
- (x) Define recursion.
- (xi) Arrangement of playing cards is an example of which sorting technique?
- (xii) Explain Ackerman function.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Differentiate between the switch and nested-if statement in C with proper example. [5]
3. Write a C program to count the number of even and odd numbers from an 1-D array. [5]
4. Write a program in C to find the area and circumference of a circle using function [5]
5. Write a C program to find the length of an array using pointer arithmetic. [5]
6. Write a C program to perform bit wise left shift operation on a number taken from the user. Write the output with an example. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Draw a flowchart to evaluate factorial of a number. [5]
(b) Write a C program to generate Fibonacci Series upto n-terms using loop. [5]
(c) Explain preprocessor directive with suitable example. [5]
8. (a) Define a recursive function to calculate factorial of a number and call this function to calculate $S = 1! + 2! + \dots + n!$, where n is user input. [8]
(b) Write a C program to swap the content of two variables using Call by Value. [7]
9. (a) Write a C program to list all the prime numbers for a given range of numbers. [8]
(b) Write a C program to check the entered character is a consonant or vowel. If it is not an alphabet return 'wrong input'. [7]
10. (a) Describe string with example. [5]
(b) How strings are represented in memory? [4]
(c) Write a C program to check the entered string is a palindrome or not. [6]
11. (a) Create a recursive C function to find the GCD of two numbers. [5]
(b) Write a C program to sort names of students in a class. [10]

*** END OF PAPER ***